

J.P. TESHAN Kavindu Jayasinghe

Immediately Available



Teshan Jayasinghe (teshankj)



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Teshan Jayasinghe

PROFILE

AI/ML and Embedded Systems researcher/engineer with experience in multimodal deep learning, computer vision, physics-guided machine learning, embedded systems, and intelligent sensing. Published research in IJDSA and CSSP, with hands-on experience developing end-to-end AI and embedded systems ranging from deep learning and signal processing to microcontrollers-based sensor integration, real-time control, and hardware-software prototyping.

EDUCATION

Year	Degree	Institute	Results
2022–2026	B.Tech in Electronics and Communication Engineering	Delhi Technological University, Delhi	7.90

B.Tech in Electronics and Communication Engineering, Delhi Technological University (2022–2026), CGPA: 7.90 (First Class). The curriculum provided a strong foundation in electronics, embedded systems, signal processing, artificial intelligence, machine learning, and computer vision, complemented by research-driven projects.

TECHNICAL SKILLS

- **Programming:** HTML, CSS, JavaScript, Python, MATLAB, C, C++
- **Data Wrangling & Math:** NumPy, Pandas, Matplotlib, Seaborn
- **Machine Learning:** Scikit-learn, XGBoost, PyTorch, TensorFlow, Keras
- **Database Management:** SQLite, PostgreSQL, SQLAlchemy, Zvec
- **Computer Vision:** OpenCV, MediaPipe, Ultralytics
- **MLops:** DVC, MLflow, Docker, Kubernetes
- **Simulation & Electronics:** LTspice, Simulink, KiCAD
- **Tools:** Jupyter, Google Colab, Git, VS Code

PROJECTS

1. Multimodal Fake News Detection System — Python, Colab

- Designed a deep learning system integrating BERT (text) and CLIP (image) with cross-modal transformer fusion.
- Implemented contrastive learning, knowledge distillation, and uncertainty-aware prediction.
- Improved robustness and interpretability of classification.

- accepted in IJDSA (2025).
 - [Interpretable and Robust Multimodal Fake News Detection with Evidence Fusion and Reliable Reasoning](#)
2. **Physics-Guided Multimodal Fault Diagnosis System — Python, Colab, LTspice**
- Developed a physics-guided neural network for analog circuit fault detection.
 - Combined signal processing features with deep learning architectures.
 - Enabled degradation estimation and robust fault identification.
 - accepted in CSSP (2026).
 - [MOCRN-Multimodal Ordinal Circuit Reliability Network](#)
3. **Hyperspectral Anomaly Detection & Injection Framework — Python, PyTorch**
- Designed a novel end-to-end framework for hyperspectral anomaly detection, integrating anomaly-class matching, data-driven anomaly injection, and cross-dataset modeling.
 - Developed a hybrid matching approach using Mahalanobis distance, Wasserstein distance, and RX score fusion, along with a realistic anomaly injection strategy combining class structure with ABU statistics.
 - Implemented cross-dataset anomaly transfer and evaluated RX, Spectral Autoencoder, and Hybrid AE+RX models.
 - Achieved near-perfect performance ($AUC \approx 1.0$) using a hybrid statistical-deep learning approach.
 - [Anomaly Class Matching via Statistical Multi Metric Framework](#)
4. **WasteVision – Computer Vision-Based Waste Detection — DVC, MLflow, Docker**
- Developed a computer vision project for automated waste identification using deep learning.
 - Implemented an image classification pipeline for recognizing waste categories to support waste sorting and recycling applications.
 - [GitHub Repository](#)
5. **BoxCrete E2E – End-to-End Concrete Strength Prediction — DVC, MLflow**
- Developed an end-to-end machine learning application for concrete compressive strength prediction using concrete mix composition and experimental parameters.
 - Built a complete workflow covering data processing, model development, and prediction integration.
 - [GitHub Repository](#)

CERTIFICATIONS

- Sensors and Actuators(NPTEL)
- [IBM Data Science](#)
- [IBM Deep Learning with PyTorch, Keras and Tensorflow](#)
- [MATLAB Onramp](#)
- [Simulink Onramp](#)
- [Statistics Onramp](#)
- [Machine Learning Onramp](#)
- [Simscape Onramp](#)

EXPERIENCE

- Researched deep learning applications for Fake News Detection.
- The paper titled "Interpretable and Robust Multimodal Fake News Detection with Evidence Fusion and Reliable Reasoning" by J. P. Teshan Kavindu Jayasinghe, Dr. Rahul Thakur, and Dr. Preeti Mehta has been accepted for publication in the International Journal of Data Science and Analytics (IJDSA) for the year 2025.

PUBLICATIONS

- J. P. Teshan Kavindu Jayasinghe, Rahul Thakur, Preeti Mehta "Interpretable and Robust Multimodal Fake News Detection with Evidence Fusion and Reliable Reasoning" International Journal of Data Science and Analytics (IJDSA), 2025 (Accepted for publication)
DOI:<https://doi.org/10.1007/s41060-025-00895-9>
- J. P. Teshan Kavindu Jayasinghe, Gayani Pamoda Kulathunga, Rahul Thakur "MOCRN: A Novel Physics-Guided Fault Modeling and Multimodal Neural Network for Fault Identification and Degradation Estimation in Analog Circuits" Circuits, Systems, and Signal Processing (CSSP), 2026 (Accepted for publication)
DOI:<https://doi.org/10.1007/s00034-026-03590-7>

POSITIONS OF RESPONSIBILITY

- Secretary of Interact Club in Nalanda College 2016
- Member of IEEE 2022
- Asia Representative of DTU International Students' Society 2025

REFERENCES

Dr. Rahul Thakur

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